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**CONCEPT SUMMARY: Logarithms**

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**Exponential Form**

**Logarithmic Form**

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**ALGEBRA**

$$b^x = y$$

$\Leftrightarrow$

$$\log_b y = x$$

**WORDS**

The base raised to the exponent  
is equal to a result.

The logarithm with a base  $b$  of  
the result (or argument) is equal  
to the exponent.

**NUMBERS**

$$3^4 = 81$$

$\Leftrightarrow$

$$\log_3 81 = 4$$

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